Finding gaps in transit access

How might we use public data to spot where transit schedules make it hard to reach essential destinations, and figure out which improvements matter most and what the trade-offs are, including first-mile/last-mile gaps?

> ***Build a working proof of concept that a municipality could realistically pilot. It can be a web app, data tool, map, dashboard or another format. A municipal leader should be able to use it without your team in the room.***

The problem

Transit access varies greatly between the neighbourhoods of a city. Some have far more access issues, and it is hard to surface where the available systems are falling short. A nearby stop may still leave people facing long waits, difficult transfers or a long walk before they can reach an essential destination. A bus stop near your home does not guarantee useful access. What matters is whether the schedule gets you to work, school, a clinic, a grocery store or childcare at the times you need, in a reasonable travel time. Coverage maps show where routes run, not whether trips are practical at 6 a.m., on Sundays or for shift workers.

The journey to and from the stop matters too. Missing sidewalks, unsafe crossings, long walks, winter conditions and a lack of bike parking or on-demand options can make a nearby stop unusable. These are first-mile and last-mile gaps.

The Organisation for Economic Co-operation and Development (OECD) and the International Transport Forum assess accessibility by the destinations people can reach within a given time. This makes frequency, connections, and walking access important alongside the location of routes and stops. [1] Statistics Canada calculates convenient access to public transit as the share of people living within 500 metres, measured along the street network, of a transit stop. In 2023, the national figure was 75.1% across 1,111 municipalities with available data, with Toronto, Montréal and Vancouver above 82%, and smaller metropolitan areas generally lower (Statistics Canada, The Daily, 2023). In 2026 Statistics Canada published a newer table that breaks access down by distance, transit carrying capacity, geography and demographic characteristics (The Daily, June 2026).

The TTC publishes scheduled routes, stops and trip times for buses, streetcars and subways. These records support comparisons across locations and times of day, but do not show actual delays, crowding or every barrier along a walking route. [2] Being near a stop does not mean a bus comes often. A tool that shows service frequency and who is affected would add something the headline statistics cannot.

How common and persistent it is

- A national study found nearly one million low-income Canadians live in urban areas with low transit accessibility, putting them at risk of "transport poverty" (Allen and Farber, 2019).
- In the Toronto area alone, the same research team estimated more than 600,000 people struggle with both low income and poor transit options.
- A Toronto study found lower-income inner suburbs had transit scores less than a third of more affluent neighbourhoods.

Questions to investigate

- How do neighbourhood density, land use and street design affect access to frequent transit?
- How do transit agencies balance frequent service on busy routes with coverage across more neighbourhoods, and who benefits from each approach?
- How long does it take residents to reach essential destinations at different times of day, including walking, waiting and transfers?
- Where do first-mile and last-mile gaps make a journey difficult, even when transit service is available nearby?
- How do transit needs differ for shift workers, caregivers, seniors, newcomers and other residents, and whose experiences might public data miss?
- Which improvements could make the greatest difference, and what trade-offs would they involve in cost, service frequency, coverage and accessibility?

What it costs

- Economic factors include limited access to jobs which narrows employment options and can force households into costly car ownership.
- Health and wellbeing in terms of missed medical appointments, harder access to fresh food, and social isolation, especially for seniors and people with disabilities.
- Equity as the burden falls most heavily on low-income households, women balancing work and childcare, and recent immigrants.

Who you are designing for

Primary users: Transit planners and transportation staff investigating gaps and comparing possible service improvements.

Secondary users: Transit-dependent residents, including students, shift workers, older adults, caregivers, and people with mobility limitations. Residents of fast-growing suburban areas and new subdivisions.

Municipal users. Equity, planning and sustainability staff who work on access to jobs, services, and housing which are all linked to mobility. City and regional councillors who need an easy-to-read evidence base.

Other stakeholders. Neighbourhood groups, accessibility advocates, destination operators and transit agencies.

What success looks like

You might build an access map, an understandable service score or a tool that compares improvements. Include first-mile and last-mile connections (the journey to the first stop and from the final stop to the destination).

- Users can see where and when access is weaker and understand the criteria behind that finding. Surfacing of specific places and times where access breaks down, not just general averages.
- The analysis considers service frequency, waiting, transfers, and access to essential destinations.

- Users can compare potential improvements and understand their benefits, resource needs, and trade-offs.
- A municipal leader can explore the working tool independently and identify a useful next step for service planning.
- A tool that shows transit access at the neighborhood level, combining walking distance to stops with service frequency, plus a way to identify the least-served areas.
- Methods that are flexible for when schedules change.
- A tool that maps how long it takes to reach essential destinations (groceries, healthcare, jobs, schools) by transit from each neighbourhood, at different times of day
- Highlights the worst gaps and who lives there
- Tests one or two "what if" improvements (more frequent service, a new route, a first/last-mile option) and shows the trade-offs
- Explains results clearly for a non-technical audience .

Constraints

- Define what "frequent", "accessible" or "underserved" means in your analysis. Show the travel times, dates and assumptions used.
- Use dates covered by the transit feed and account for service calendars. Scheduled service is not the same as reliable service.
- Make walking estimates and accessibility gaps visible. A straight-line connection does not establish a safe or wheelchair-accessible route.
- Do not generalize from a few sample locations to all residents. Population or equity claims need supporting geographic and demographic evidence.

Public datasets to explore

You may use other public data that better supports your solution. Record your sources and dates, and make data limitations visible.

1. **Merged GTFS TTC Routes and Schedules (City of Toronto data and downloads)**

   Includes routes, stops, trips, stop times and service calendars. Use it to measure departures, scheduled connections and travel times. Join tables using route, trip, stop and service identifiers, and check the feed's service period.

2. **Library Branch General Information (City of Toronto data and downloads)**

   Includes branch locations, coordinates, opening hours and service features. Libraries provide a useful example of essential community destinations. Select physical branches and check opening hours; this dataset does not cover all essential services.

3. **Grand River Transit GTFS feed** (Region of Waterloo open data; GTFS schedule, with route and stop data). A local, well-maintained feed. The feed is also listed on the Mobility Database, and teams can swap in any city's GTFS feed.
4. **Statistics Canada, Table 23-10-0314-01: Access to public transport by distance and carrying capacity, geography, gender, selected demographic and socio-economic characteristics, and count of public transport stops** (see the June 2026 release). Useful as a benchmark and for demographic context.

Explain your thinking

Show who the tool serves, why you designed it this way and how you checked its outputs. Include clear instructions, source links and any assumptions. Explain what a municipality would need to test or add before piloting it. If you use AI, make its role and limitations clear.

Background sources

[1] OECD and International Transport Forum. Benchmarking Accessibility in Cities

[2] Toronto Transit Commission. Merged GTFS TTC Routes and Schedules